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# How Do Global Shocks Affect Australia?

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A summary deck prepared from the published paper

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# Small, open and fully exposed — how much does it actually matter?

Australia is a small open economy with open capital markets. Global shocks reach it quickly, and policy must decide how much weight to give them.

The evidence disagrees: foreign shocks have been estimated to explain 20 to over 50 per cent of Australian GDP variation, and 10 to 70 per cent of the exchange rate.

And the channel is contested: trade, the exchange rate, asset prices, or global banks.

**WHY IT MATTERS**

How much weight to put on global developments.

Where monitoring effort should sit.

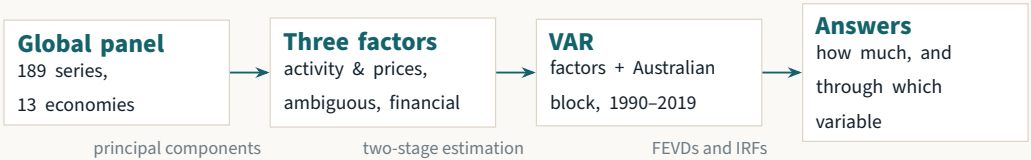
Whether policy faces the same shock twice.

# A FAVAR lets the data define “global”, not assume it means “American”.

Most Australian studies put US variables in the foreign block. That misses shocks which reach Australia without passing through the United States — an Asian financial crisis, for instance.

| | | |
|---|---|---|
| 01 | **Spans the shocks** | 189 series, 13 economies, compressed into three factors. |
| 02 | **Measures importance** | Variance decompositions for the Australian block. |
| 03 | **Locates the channel** | Adds financial variables, then tests whether banks respond. |
| 04 | **Stress-tests the answer** | Against a US-only VAR, 116 other factors, and China. |

# Two stages: extract global factors, then estimate a small VAR.




**Australian block** CPI inflation, the cash rate, unemployment, the TWI and real GDP.

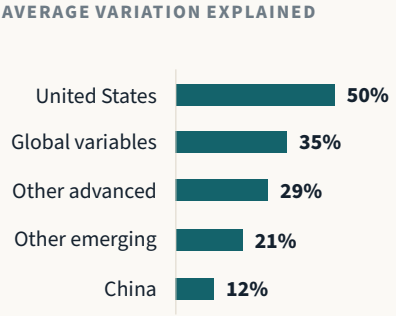
**Small open economy** Australian variables cannot move the global factors, on impact or with a lag.

**Sample** Quarterly 1990–2019, excluding COVID. The factors are not structural.

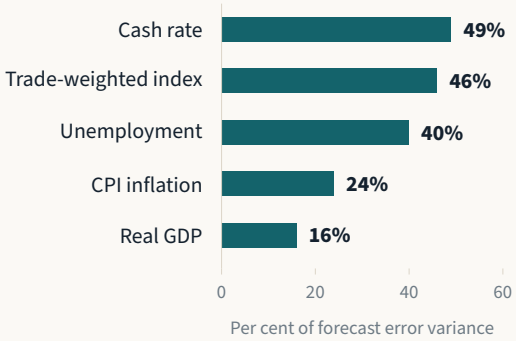
# Three factors capture about 30 per cent of the global panel — very unevenly.

1. **Activity and prices**
   US prices load most: 41 per cent of their variation.
2. **Not clear-cut**
   Broadly even across categories.
3. **Financial conditions**
   Explains financial variables most.

The factors are orthogonal and carry no economic label: they summarise common variation, not identified shocks. Three were chosen by the Bai–Ng criteria.



# Global shocks explain half of the cash rate and the exchange rate, a sixth of GDP.



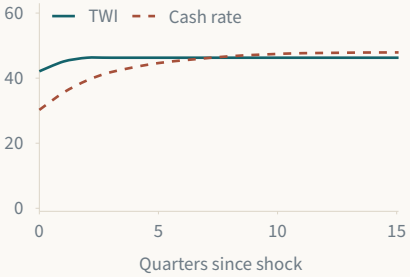

Global shocks dominate the two variables that are supposed to absorb them, and matter much less for output.

The gap between the exchange rate and GDP is the paper's central quantitative result — and the reason the next question, through what, matters.

# That pattern is what a buffered economy looks like.

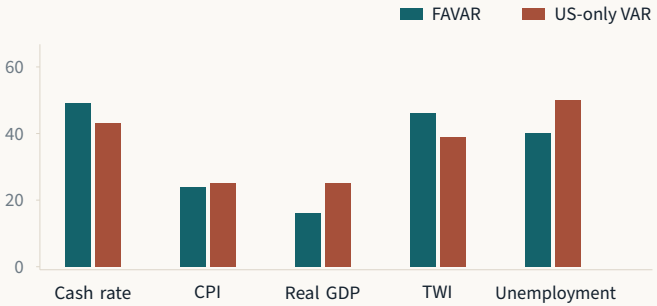
**The dollar moves first.**
The TWI reprices within a quarter and stays there, as efficient markets would predict. It takes the shock so domestic activity does not have to.

**The cash rate moves later.**
It responds gradually, consistent with the RBA updating its outlook slowly or adjusting in steps. Either way, policy offsets the shock rather than passing it on.




**What is left is composition, not collapse.** GDP held up through the 2001 US recession and the GFC while unemployment still rose, and the pain was regional: mining-exposed Western Australia and Queensland held up while New South Wales deteriorated.

# A US-only VAR does nearly as well.




Per cent of forecast error variance

The FAVAR explains more of the cash rate and the TWI, about the same of unemployment, and less of CPI and GDP.

None of the differences is statistically significant.

Over 1990–2019, US variables were effectively a sufficient statistic for the global shocks that mattered here.

# A fourth factor adds little — and the best ones are Chinese.

Principal components explain the global panel, not Australia. So the paper swaps the fourth component for each of 116 alternatives.

01 **Nothing beats the baseline**

No alternative factor adds a material share.

02 **The best are Chinese**

They load on Chinese variables, not the global panel.

03 **But China alone explains less**

Chinese variables alone do worse than the three factors.

**THREE EXPLANATIONS**

China may be insulating itself from global shocks.

Chinese shocks may show up in global data before they do in Chinese data.

Chinese demand may matter for the level of Australian activity, not its quarterly variation.

# Asset prices take the shock. Bank balance sheets barely move.

**ASSET PRICES**

55%

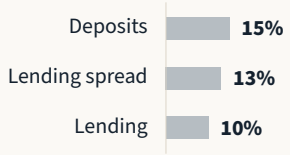
of the variation in the ASX all ordinaries

40%

of the variation in housing prices

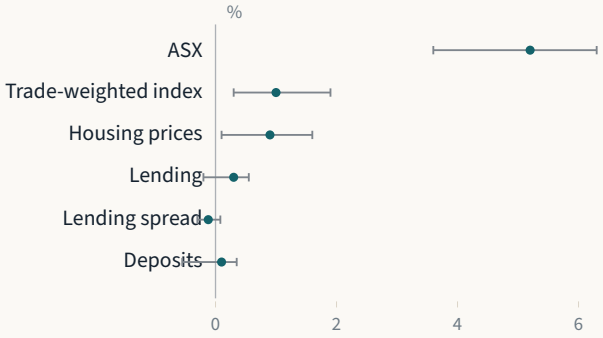
Stock prices adjust on impact; housing prices build up over time, consistent with a slow-moving market or an effect intermediated by the cash rate.

**BANK BALANCE SHEETS**




Around a tenth to a sixth of the variation, with confidence bands that span zero. If banks were the channel, these are the variables that should move.

# Shock the financial factor: prices move, bank quantities do not.




A one standard deviation shock to the factor that loads on financial variables lifts the ASX by about 5 per cent and the TWI by about 1 per cent.

Lending, deposits and spreads end the horizon close to zero, with bands that straddle it.

That is what a channel that is *not* operating looks like.

# Why Australian banks are insulated: three structural reasons.

**ASSETS**

## Lending, not trading.

Australian banks hold fewer marketable securities and earn less from trading than global peers. More of their assets are loans, largely home lending.

**FUNDING**

## Offshore, but hedged.

Offshore bonds are just under 15 per cent of funding, and about 90 per cent of those liabilities are hedged. The sector runs a net foreign-currency asset position.

**STRUCTURE**

## Concentrated, profitable.

The Big 4 dominate and foreign banks are marginal. High profits and low leverage make forced deleveraging less likely.

**The counter-argument.** Concentration can also amplify a shock concentrated in one major bank — a too-big-to-fail dynamic rather than a buffer.

# The channels that do transmit look traditional.

| CHANNEL | STRENGTH | EVIDENCE |
|---|---|---|
| **Exchange rate** |  | About 46 per cent of TWI variation, within a quarter. |
| **Asset prices** |  | 55 per cent for the ASX, 40 per cent for housing. |
| **Trade** |  | About 30 per cent for the terms of trade. |
| **Banking** |  | 10–15 per cent; the bands span zero. |

The exchange rate and the cash rate do most of the buffering — which is why the shocks show up in prices rather than quantities.

# The banking channel is a feature of a financial system, not a law.

## FOR POLICY

### Monitor the right link.

If spillovers arrive through the exchange rate, trade and asset prices, that is where scenario work belongs.

### Do not import another country's channel.

The global-banking result is about internationalised, trading-heavy systems.

### Structure is endogenous.

The features that muted the channel here are shaped by regulation. Change them and the answer can change.

## FOR RESEARCH

### A selection problem.

Spillovers are identified from heterogeneity in foreign exposures, so the countries that can be estimated may be exactly where the channel is strongest.

### Test it where the structure differs.

Systems like Australia's are the useful case: an absent channel where theory says it should be weak is evidence about the channel itself.

# What this design cannot say — and what it does.

## LIMITS

**The sample stops in 2019.** A linear framework cannot absorb COVID, and the paper does not claim the 1990–2019 mix of shocks will repeat.

**The factors are not structural.** They identify that a shock is global, not which shock it is.

**No long-run identification.** China may matter for levels in a way quarterly variation cannot reveal.

**One country, one structure.** This is evidence about Australia, and a prompt to compare banking systems — not a theorem about small open economies.

## THREE TAKEAWAYS

1. Global shocks explain about half of the variation in the cash rate and the exchange rate, but only about a sixth of real GDP.
2. They do not reach Australia through the banking system: lending, deposits and spreads do not respond.
3. That is a statement about the structure of Australian banking, not about small open economies in general.

SOURCE

# The paper.

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IN ONE LINE

Global shocks are large, but Australia’s exchange rate and cash rate absorb most of them — and its banks are not the pipe.

Charts are redrawn from the paper’s Figures 3, 5, 7, 11 and 12, with values read from the published figures and rounded. The paper and its figures are published by the Reserve Bank of Australia under a Creative Commons Attribution 4.0 International licence; the views are the authors’ and not those of the RBA.